

AI-Powered Student Study Planner

Project Description

Overview

A mobile application for managing academic preparation across subjects and deadlines, dynamically generating personalized study routines using AI algorithms.

Problem Statement

Traditional study planning relying on manual schedules or physical planners is time-consuming and static. When students encounter unexpected delays or shifting exam dates, updating physical planners manually is inefficient. This results in disorganized preparation, missed topics, and exam stress.

Proposed Solution

AI-Powered Student Study Planner is built as a Flutter mobile application that takes subject difficulty, task deadlines, and daily student availability into account. It uses an AI engine to calculate dynamic study blocks, providing students with a structured calendar view and real-time progress updates.

Key Features

#	Feature	Purpose
1	User Login & Onboarding	Secure authentication and setup of daily study time limits
2	Subject & Difficulty Manager	Organize courses and assign difficulty weightage (1–5)
3	Task & Exam Tracker	Record assignments, tests, and estimated prep hours
4	AI Routine Generator	Process constraints and auto-generate study allocations
5	Interactive Calendar View	Daily time blocks with completion toggles & status
6	Dashboard & Progress Tracking	Real-time stats, deadlines, and task progress

Technology Stack

- Flutter and Dart for cross-platform app implementation
- Material Design 3 for responsive UI architecture
- Firebase Authentication & Cloud Firestore for data management
- Google Generative AI (Gemini API) for backend schedule generation

Expected Impact

The application optimizes student time management, reduces routine creation overhead, prevents exam preparation bottlenecks, and improves academic output through structured, adaptive schedules.

Target Users

High school, undergraduate, and postgraduate students balancing multiple academic subjects, assignments, and exam deadlines simultaneously.